Jiwoon Choi

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EDUCATION

UNIVERSITY OF PENNSYLVANIA — **Philadelphia, PA**
Master of Applied Science, Computer Science — Expected: May 2028
- Coursework: Introduction to Software Development (Java), Discrete Math, Linear Algebra, Introduction to Computer Systems

KOREA MILITARY ACADEMY — **Seoul, Korea**
B.A., Military English & B.M.A.S., Military Art and Science — Mar. 2016 – Mar. 2020
- Honors: Highest Distinction (1st in major); Dean's Award for Academic Excellence
- GPA: 3.65 (U.S. equivalent)
- Relevant Coursework (all A): Introduction to Computer Science, Probability & Statistics, Unmanned Military Robot (Machine Learning), Defense Modeling & Simulation

SEOUL, SEJONG & KYUNG HEE CYBER UNIVERSITIES — **Seoul, Korea**
- Coursework (all A/A+): Data Structures & Algorithms, Object-Oriented Programming (C++/C#), Systems Programming (Linux), Java, C, Discrete Math, Linear Algebra — Sep. 2023 – Jun. 2024

SKILLS

- Languages: Python, C++, Java, C
- Concepts: Data Structures & Algorithms, Object-Oriented Programming, Software Development Life Cycle
- Developer Tools: Git/GitHub, Linux, AI-assisted development (ChatGPT/Codex), ROS 2, Raspberry Pi
- ML/Data: TensorFlow, scikit-learn, NumPy, Pandas
- Certifications: SQL Developer (SQLD), ADsP — Korea Data Agency

PROJECTS

Collaborative Robot Assistant (CRA) — Ongoing | ROS 2, Python, Gazebo
- Develop a modular ROS 2 software system that receives, validates, queues, and executes user commands across multiple components for a simulated mobile robotic arm
- Design and test state-based command workflows with real-time status tracking, error handling, and execution feedback
- Implement and debug inter-process communication between software modules using ROS 2 topics, validating component behavior through simulation-based testing

Voice-Based Weather Assistant | Raspberry Pi, Python, NLP | GitHub
- Built a voice assistant that parses natural-language queries, retrieves weather data via API, and responds using TTS
- Implemented rule-based NLP, time reasoning, and an ASR→NLP→API→TTS pipeline; created a demo

Voice-Controlled RGB LED System | Raspberry Pi, Python | GitHub
- Designed a real-time voice-operated LED controller using Google Speech Recognition and Raspberry Pi
- Implemented audio processing and event-driven hardware control; published code and demo video on GitHub

EXPERIENCE

REPUBLIC OF KOREA ARMY — Mar. 2020 – Feb. 2026
Infantry Officer (Captain)
Selected roles: Liaison Officer, Executive Officer, Guard Post Commander, Platoon Leader
- Served as primary liaison between Korean Army and three U.S. Army units, coordinating cross-unit operations
- Led organizations of 100+ personnel across operations, logistics, personnel, and cross-functional teams
- Commanded 24/7 DMZ border operations in coordination with the United Nations Command
- Improved automated surveillance by redesigning PTZ camera configurations using geographic and operational-pattern analysis, reducing false alarm rates from 78% to 33%
- Developed joint Korea-U.S. Army professional development initiatives to improve interoperability

VOLUNTEERING / TEACHING EXPERIENCE

Youth Mentor | ChildFund Korea — May. 2023 – Sep. 2023
- Mentored and tutored an underprivileged student in reading and mathematics.
- Guided the student in academic planning, career exploration, and long-term educational preparation.